Assignment - 1

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Section: 7

Fall 2025 set A

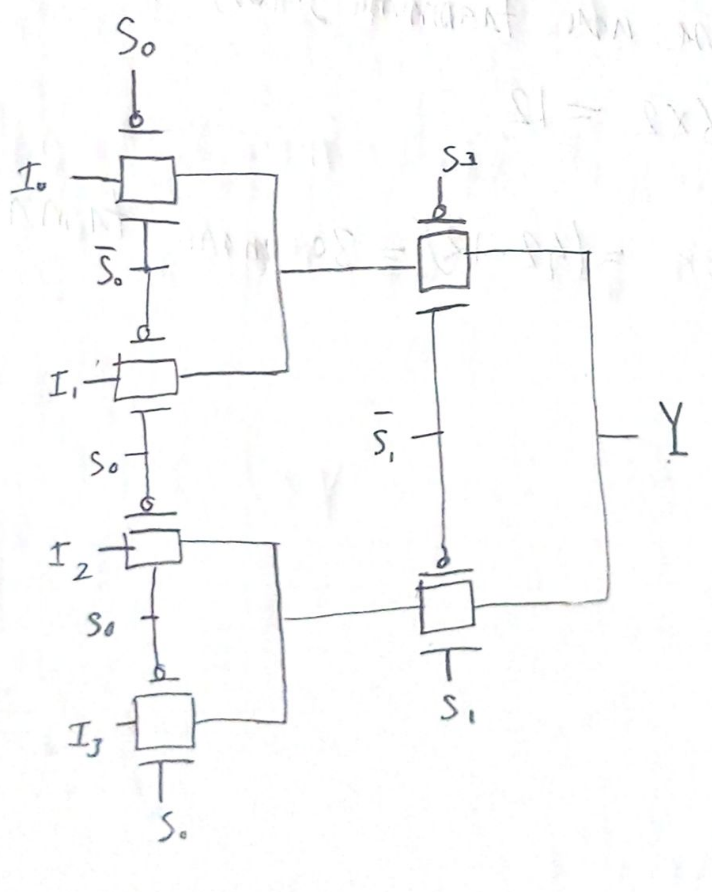
Ans to the Q No-3

(a) (i)

S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

logic function, $Y = \bar{S_1}\bar{S_0} I_0 + \bar{S_1} S_0 I_1 + S_1 \bar{S_0} I_2 + S_1 S_0 I_3$

(ii)



(iii) From (i) $Y = \bar{S_1}\bar{S_0}I_0 + \bar{S_1}S_0I_1 + S_1\bar{S_0}I_2 + S_1S_0I_3$

We know

1 AND operation 8 transistor needed (3NMOS + 3PMOS + 2 inverter)

~~x~~ OR operation 8 transistor needed

Now for 4 AND terms $= 8 \times 4 = 32$.

$\therefore$ number of transistors $= 32 + 8 = 40$.

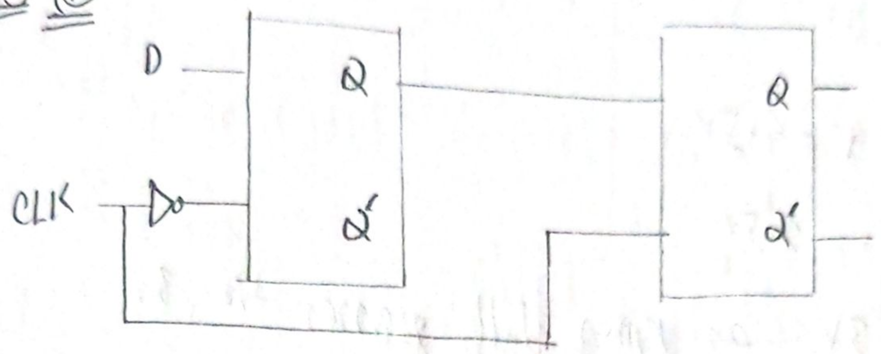
For the final inverter,

Total number of transistor $= 40 + 2 = 42$.

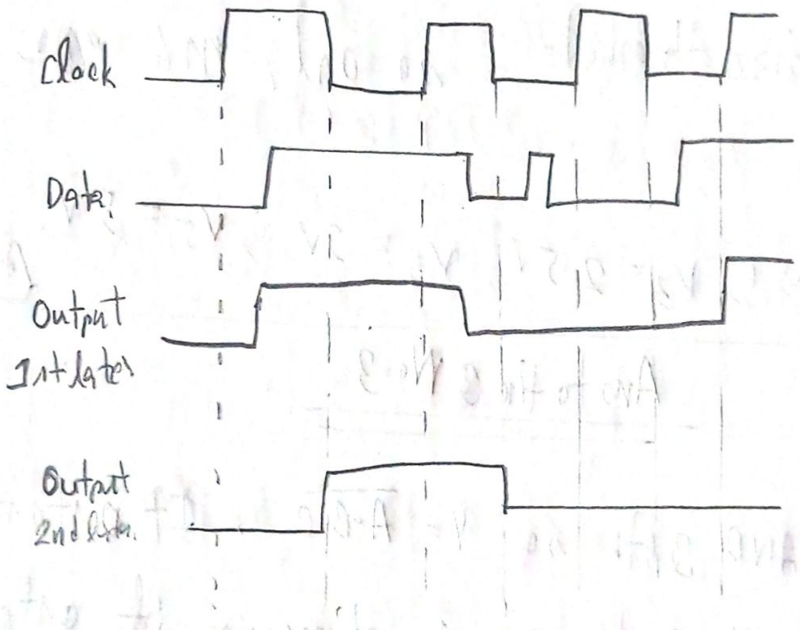
From (ii) we see use transmission gates, so number of transistor $= 6 \times 2 = 12$.

So in CMOS logic $= (42 - 12) = 30$ more transistors are needed.

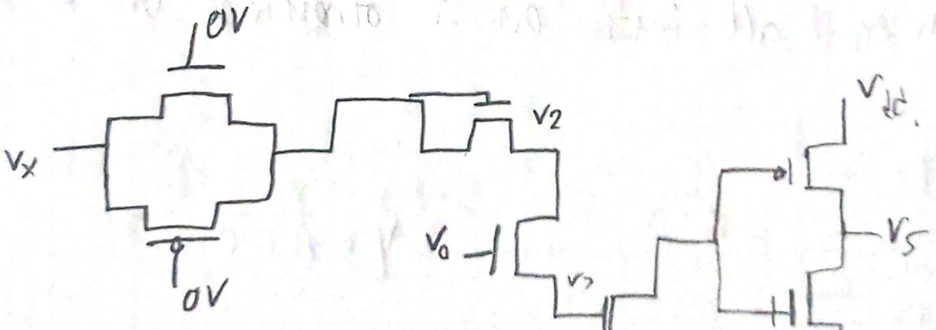
(b) (i)



(ii)



(c)



$V_x = 5V$, $V_y = 3V$, $V_2 = 3V$, $V_{dd} = 5V$, $V_{in} = 0.5V$ $|V_{tn}| = 0.3V$

Here,

$V_1 = 5V = V_X$.

$V_2 = (5-0.5)\ V = 4.5V$,

$V_3 = (3-0.5)\ V = 2.5V$,

$V_4 = V_Z = 3V$. as pmos will passes the 3V

$V_5 = V_{dd} = 5V$ because M7 is off $V_G - V_S = 3-3 = 0V$, which is less than threshold so only M6 pmos.

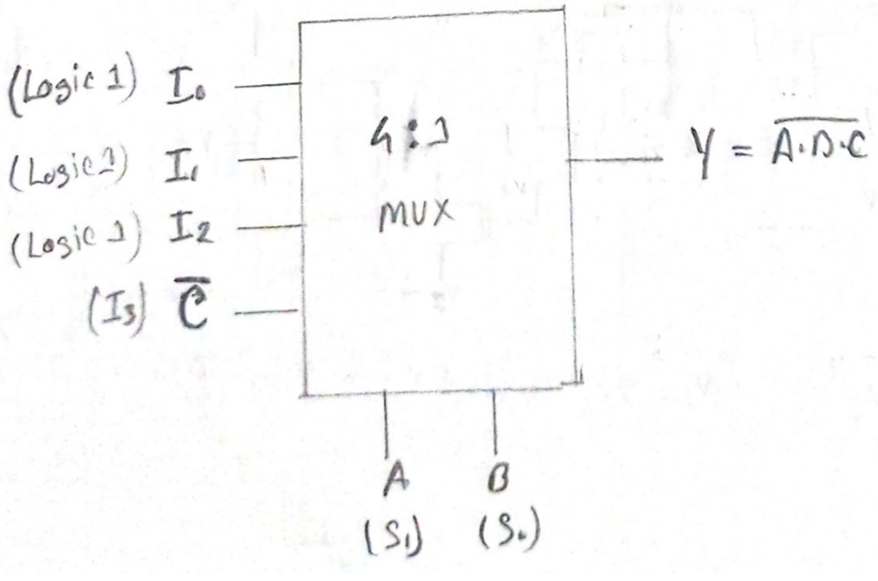
So,
Node voltages,

$V_1 = 5V$, $V_2 = 4.5V$, $V_3 = 2.5V$, $V_4 = 3V$, $V_5 = 5V$. Ans

Spring 25

Ans to the Q No-3

(a) As 3 input NAND gate so $Y = \overline{A \cdot B \cdot C}$. It outputs a 0 only when all inputs are 1 otherwise it outputs 1.

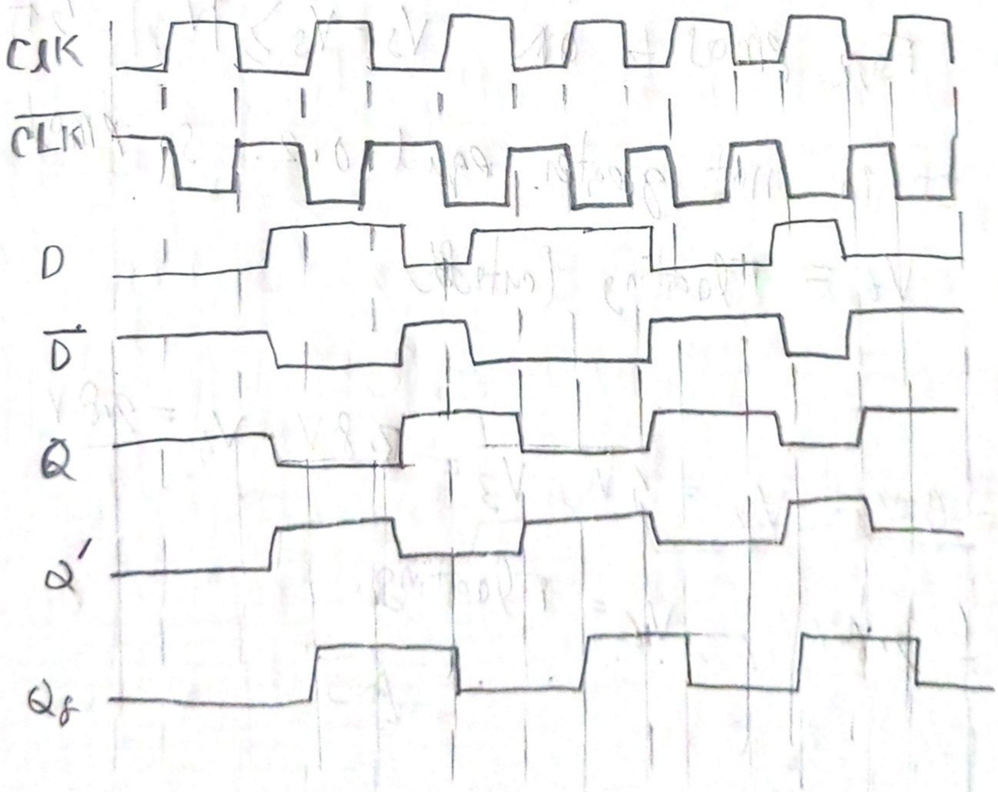


(b) Device 1 is a Positive Level D Latch. Here the input ~~transm~~ transmission gate has it's nmos ~~conne~~ connected to CLK and pmos to $\overline{CLK}$ meaning it passes data when CLK = 1 and the feedback loop is enabled when clk = 0.

characteristic table:

clk	D	Q(t+1)
0	X	Q(t) (Hold)
1	0	0
1	1	1

(c)



This is a positive edge triggered d flip flop.

(d) Here, $V_{DD} = 4V$, $V_{in} = 0.2$ and $|V_{tp}| = 0.2V$.

As input signal passing through transmission gate is 0V.

So, $V_1 = 0V$.

For V_2 gate voltage 0V, the NMOS is cutoff and PMOS is ON. so $V_2 = V_{DD} = 4V$.

For $V_3 = (4-0.2)V = 3.8V$,

$V_4 = (3 - 0.2)\ V = 2.8V$.

$V_5 = 2.8 - 0.2 = 2.6V$.

Now For pmos to ON $V_S - V_G \geq |V_{tp}|$ but $(2.6-4) = -1.4$ so it is not greater equal 0.2. So pmos will be off and V_6 = Floating (cutoff)

So,

$V_1 = 0V$, $V_2 = 4V$, $V_3 = 3.8V$, $V_4 = 2.8V$

$V_5 = 2.6V$, V_6 = Floating.

Ans

Summer-25

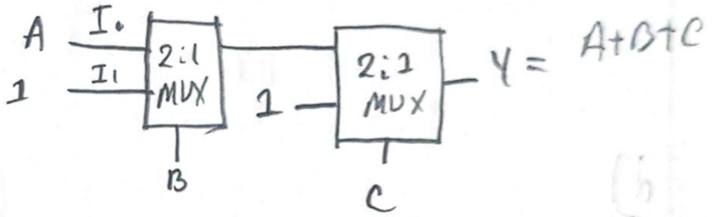
Ans to the Q No-3

(a)

The scenario 1 is a 3 iput or gate. The equation is $Y = A+B+C$.

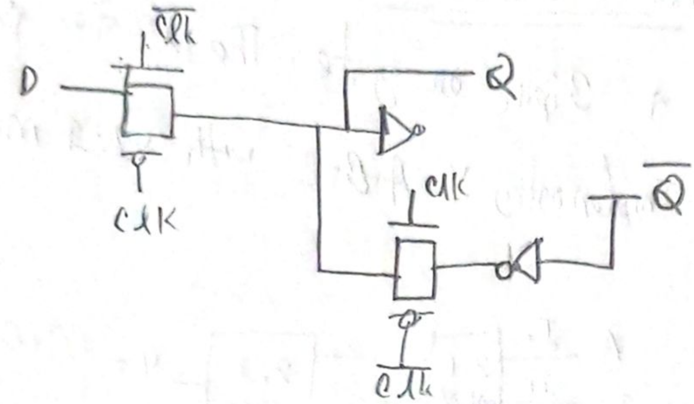
Implementing $Y = A+B+C$ with 2:1 mux

A	B	C	Y
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1



(b) For scenario 2 the circuit in figure 1 is a D flipflop. In figure 2 it can bee seen that the Qd is copying data when the clock is low, so the D flipflop is a negative edge trigged flipflop and the device in figure 1 is a D latch. Two latches are used to build the d flipflop that copies and hold data according to clock cycle. The latch here is a negative level triggerd latch as it copied D when clk is low.

(c) Device 1 is a D latch. Now using transmission gates



(d)

Nmos transistor can produce good 0 but bad 1. NMOS ~~eq~~ equation is $V_{GS} \geq V_T$. So for it to turn ~~on~~ ON V_{GS} has to be greater than V_T ~~...~~. So $V_S = V_G - V_T$. So we always get a less amount of voltage from the one provided. Thus it produces bad 1 and the overall voltage can be perfectly ~~zero~~ 0 which means transistor is off so CMOS can produce good ~~zero~~ 0 but bad 1.

On the other hand PMOS transistor can produce good 1 but bad 0. In ~~....~~ case of PMOS. $V_{SG} \geq |V_T|$, so $V_S = V_G + V_T$. So even if $V_G = 0$, $V_S = V_T$. As a result V_S can never be a proper 0. So it will always have amout voltage. But ~~if~~ id V_G is given, proper amount of voltage V_{SG} becomes .. bigger as we are adding V_T.. Thus PMOS produces good 1 but a bad 0.